A Self-Closing Gate Conundrum

#1 Victory Royale

Element K: Reflection on the Design

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

The problem statement is "student G's family needs a fence latch that closes the gate on its own because his family does not want unwanted animals and people inside their fence because it is a safety concern." When we brainstormed we wrote down as many ideas as we could in five minutes. We then chose the top five best ideas from all of the ideas that our group came up with and used the decision matrix to find the best idea. We sketched a design for our prototype and our group agreed that our sketch was good. We built it using materials that were in Ms. Z's classroom. Our design met two of our criteria and what that showed us was that we needed to add a cushion in between the clamp and the fence. We were not able to do any tests on this because it was more for us to get a feel for the design. We made the second prototype which was made from materials that Ms. Z bought us. We ran all of our test procedures and all of the tests that we ran fit our criteria and constraints.

When we brainstormed we came up with a lot of ideas as a whole group. We had a few disagreements on what should be our top five ideas but we compromised. When we were using the decision matrix there was little bit of confusion on what a tension spring is, two members of our team thought that a tension spring was a compression spring which was just a simple misunderstanding that we resolved and it only wasted a little bit of time. For our prototype design we already knew what we wanted to do so it was quick and easy. Building the second prototype had a few problems like not being able to find a way for it to not slide down and scratch the fence. Testing we had a few problems like how the member of our team who was meant to test did not do it for many days. For iteration we had no problems, we just knew that we needed to add a strip of cloth to protect the fence from getting scratched.

Make sure that your whole team understands what you are making before you begin your process. If a problem comes up and needs to be fixed, figure out what the best solution is before trying several ideas.